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Designing automatic card dispensers based on design thinking approach and selecting the suitable alternative

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Abstract

The “Historical Game Journey (HGJ) Project,” which aims to transfer cultural heritage to children through gamification, plans to deploy kiosk stations named “Information and Communication Station (ICOS)” at historical sites. Within the ICOS, there is an “Automatic Card Dispenser (ACD)” providing information cards about the historical buildings, and it constitutes a significant cost item in the project budget. In this study, with the aim of saving costs and staying within the budget, the ACD is intended to be developed instead of being purchased as a commercial off-the-shelf (COTS) product. The aim of the study is to design and develop an ACD prototype that optimally meets functional requirements and offers greater cost efficiency, specifically for implementation within the ICOS system. In this context, three alternative ACDs—with similar technologies and software but different operating principles—were designed using the Design Thinking approach, and the most suitable one was selected using the Weighted Evaluation Criteria System (WECS) method. In this way, the cost ratio of the ACD within the project budget was reduced from 21.8% to 2.42%. The developed ACD can be used practically in applications such as card games, educational kiosks and ticket vending machines used in parking lots, public transportation, highways etc. with simple adaptations.

Keywords: Card dispenser with servo motor, Arduino controlled card dispenser, Historical Game Journey, Decision making, Dispensing machine, Vending machine, Card dealer

Introduction

Raising awareness of cultural heritage within society plays a crucial role in shaping urban identities, sustaining national continuity, preserving national values, and protecting global cultural heritage [10]. Instilling such awareness in adults—whose perspectives have often reached a point of saturation—is a lengthy and demanding endeavor. Therefore, it is widely considered more effective to cultivate cultural heritage awareness in children, whose minds are still clean, pure, and untainted. Every child begins life as a natural researcher: filled with questions, eager to learn, and driven by curiosity [17].

Children must be aware of their cultural heritage to protect them. In other words, children should be taught their history and cultural heritage. Only then they are expected to protect these important values. Gamification method can be used in teaching cultural heritage. Children learn many things while playing. In time, forms, methods and boundaries of the games have changed. However, human beings continued to play in accordance with their current conditions regardless of time and place [38]. The “Historical Game Journey (HGI)” project aims to teach children about their historical and cultural heritage at their young age by playing games [13]. In the project, a smart scepter (which a child will carry around during his/her journey in the historical site), Information and Communication Station (ICOS) (which is a specially designed kiosk that will be temporarily placed next to each historical building determined in the area), information card (which is given to the child from ICOS after touching the scepter), and an Epic Journey Scrapbook [11] (which is given to the child at the end of the journey and includes a story in which the hero is the child with the child’s photos in front of the historical building) will constitute the basic elements of the Project [12].

The motivation for this study is to reduce the budget required for the costly “Card Dispenser Mechanism” which is embedded within ICOS and delivers the information card to the participant after the smart scepter is touched. Due to the unexpected increase in exchange rates between the project proposal and acceptance phases, the HGI project must be completed with a more limited budget than initially planned. Therefore, it is necessary to cut costs on high-expense items in the project. This study aims to design an “Automatic Card Dispenser (ACD)” and develop an ACD prototype for the ICOS. The design scheme of the ICOS with the ACD is presented in Fig. 1. The RFID-based ACD, identified during the HGI project proposal phase and considered one of the main components of the project as well as a critical part of the ICOS, is among the highest-cost

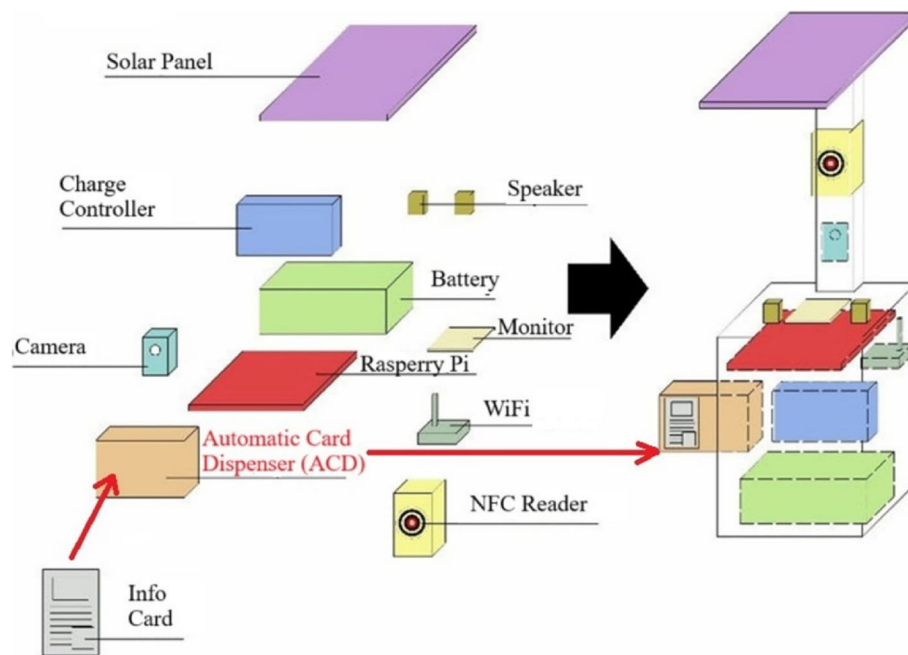


Fig. 1 Design scheme of the ICOS (Resource: [14], p.4)

items if procured from the market. For this reason, the goal was to develop the ACD as a new product within the scope of the capabilities of the personnel involved in the project rather than procuring it from the market, and to remain within the constraints of the limited project budget. In this context, three alternative ACD prototypes based on “Arduino” and “Servo Motor” were first designed, then manufactured using 3D printers, and finally, the most suitable one for the project was selected. The technologies used in the ACD were chosen based on criteria such as cost-effectiveness, accessibility, functionality, widespread application, and responsiveness to requirements. The technologies planned for the ACD production are briefly explained below.

Among the technologies considered for ACD development, Arduino stands out as a versatile and programmable platform for signal detection and actuator control. Arduino is a microcontroller board that can detect the electrical signals from the buttons and/or sensors and enables the actuators that provide light (led), sound (buzzer), motion (motor) to be operated in the desired direction thanks to the software loaded into it [22]. Microcontroller is an integrated, programmable, low-cost and small-sized element that contains memory, input–output and microprocessor units [7]. Arduino Uno has 14 digital input/output pins, 6 analog inputs, 16 MHz crystal oscillator, USB connection, power input, ICSP header that allows programming on the circuit, reset button, serial-to-USB converter and Atmega328 microcontroller as shown in Fig. 2. It is a platform that can both communicate and be programmed over the port and contains the necessary components to support the microcontroller [7, 29]. The microcontroller on the Arduino board is programmed with the wiring-based Arduino programming language and this program is uploaded to the board with the help of the “Processing” based “Arduino Software Development Environment” [22]. Arduino board is an open-source board, easily programmable and cheaper than microcontrollers on the market.

Another key component in the ACD design is the servo motor, which enables precise directional control essential for card dispensing mechanisms. Servo motor is one of the feedback DC type motors used in many system applications where up-down or right-left movement needs to be controlled and is extremely common in robotics. Servo motor provides high torque at low speed. There are three wires in a servo



Fig. 2 Arduino Uno electronics card

motor. One is used for powering the motor, the other to ground the motor, and the last one is to send a position control signal to the motor. The servo motor is controlled by a series of pulses. The length of this pulse determines the position that the servo will take [16]. Commercially available servo motors—illustrated in Fig. 3—are available in various models capable of rotating either 180 or 360 degrees.

In terms of physical prototyping, 3D printing technology was employed to fabricate the structural components of the ACD with high precision and efficiency. 3D printers are new age production tools that work with the logic of a laser or inkjet printer and transform products from digital media into a solid three-dimensional object in a layer-based structure defined as “additive manufacturing.” It defines the whole product as chopped pieces and produces a part of it at every stage. For example, a freely movable bearing with its balls can be produced as a single piece using a 3D-printer. Additive manufacturing, in oppose to the subtractive manufacturing process, allows production without resorting to any cutting, drilling, or sanding processes. In this way, it becomes very easy to produce even complex objects [6].

The card dispensers to be designed in the context of this study are generally integrated into machines commonly referred to as “dispensing machines” or “vending machines.” For consistency, the term dispensing machine will be used in this study. Kumar ([19], p.60) defines a dispensing machine as an automated device that offers various products such as snacks, beverages, tickets, notebooks, or cards to users. The products are dispensed after the user inserts cash, a credit card, or a specialized card into the machine. Similarly, Baum ([3], p.257) describes dispensing machines as devices consisting of two mechanisms: a card reader and a candy dispenser. Special



Fig. 3 Servo motor

cards are inserted into the card reader, triggering the release of the appropriate number of candies.

A preliminary review of the literature conducted across leading academic databases—including Scopus, Web of Science, ProQuest, ScienceDirect, and Google Scholar—reveals a limited number of scholarly studies that explicitly examine the design of “card dispensing machines” or “card vending machines.” The existing body of research predominantly addresses dispensing systems developed for a variety of application domains, such as fast food [20], personal protective equipment [31], snacks [26], face masks [9], additive manufacturing [25], pharmaceuticals [4], ticketing [34, 36], and card-based systems [37]. These studies tend to focus either on the holistic design of dispensing technologies or on the enhancement of specific subsystems. Another preliminary literature review focusing on the design of robotic card dealers identifies several contributions related to automated card distribution in casino and recreational contexts. However, no existing study specifically addresses the mechanical or functional design of a dedicated card dispensing mechanism. While selected studies on card and ticket dispensing systems, as well as robotic card dealer design, are summarized in Table 1, information regarding the current study is presented in the final row.

Additionally, several tool design studies employing systematic and user-centered approaches—such as those by Mutlu et al. [27], who evaluated screw-type non-contact sealing elements for high-speed applications using functional decomposition and weighted criteria; Mascarenhas et al. [23], who developed a high-temperature valve testing apparatus through multi-sample experimental setups, Parlar et al. [30], who designed an automatic food mixer by applying design thinking and user observation, Kaya and Erden [18], who created a 48 V DC emergency LED lamp for telecom cabinets using a case-based prototyping approach, and Manavis et al. [24], who explored wearable jewelry design through a holistic methodology combining empathy mapping, creative ideation, and CAD modeling—demonstrate the application of design thinking, evaluation criteria, and prototyping across diverse domains. However, also none of these studies specifically address the mechanical design of a dedicated card dispensing mechanism. The current study offers a unique contribution to the literature by designing and developing a card dispensing mechanism and proposing comparative evaluation criteria for researchers working in this domain. The subsequent sections of the study present the research methodology, the development phases of the alternative products, the selection of the most suitable alternative, the evaluation of the products, and conclude with a summary of findings and final remarks.

Method

The RFID-based ACD, which is one of the main elements of the project and main part of ICOS, is one of the highest cost items in the project if it is procured from the market. For this reason, the aim of the study was to design an ACD and develop a prototype that best meets functional needs while offering more cost-effective solution. To achieve this, the following process was implemented:

Table 1 Summary of selected studies

ID	Study	Aim	Method	Application	Key contribution
1	Sandnes et al. [34]	To evaluate the user interface of the Taiwan High Speed Rail ticket vending machine using principles from Human-Computer Interaction (HCI)	Expert evaluation based on 16 design heuristics derived from HCI literature	Public ticket vending kiosks with touchscreen interfaces, specifically those used by Taiwan High Speed Rail (THSR)	The study proposes practical and context-specific design heuristics for public kiosks, identifying usability flaws systematically
2	Siebenhandl et al. [36]	To develop a user-centered design approach to improve the usability and user satisfaction of self-service ticketing machines	Applying an iterative design process including user needs analysis, prototyping, and usability testing	Self-service kiosks and ticket vending machines for public transportation	Providing a more accessible and intuitive ticket machine interface developed through user-centered methods
3	Shitbashi [37]	To develop a gate mechanism for an automatic ticketing system that operates without user intervention	Applying an engineering-based system design involving RFID technology, sensor integration, and access control algorithms	Contactless card-operated ticket gates used in public transportation systems in Japan	Presenting an autonomous ticketing system that minimizes user interaction while ensuring secure and rapid access
4	Shippy et al. [35]	To develop an initial design for a robotic system that automatically deals cards in poker games	Creating a conceptual prototype involving mechanical and electronic components, guided by design thinking to analyze user needs	Robotic systems that automate physical card dealing at poker tables	Presenting a user-centered robotic dealer concept with demonstrated technical feasibility and potential for future manufacturability
5	Ljungberg and Hardell [21]	To design an automated card dealing system and enable its operation through mechatronic principles	Conducting a practical engineering process involving mechanical design, electronic control, and software integration	Mechatronic systems that automatically distribute physical cards for use in card games	Developing a functional and manufacturable automated dealing system as a real-world application in mechatronics
6	Halaç et al. (Current Study)	To design and develop a cost-effective automatic card dispenser mechanism for ICOS kiosks used in the HGJ project, and to select the most suitable alternative through comparative evaluation	Applying the Design Thinking approach and SCOPES framework; developing three Arduino- and servo motor-based prototypes; selecting the optimal design using the WECS	ICOS in the HGJ project; potential adaptation for ticketing and gaming kiosks in public settings	Presents a novel mechanical card dispensing solutions with original software; introduces a multi-criteria evaluation framework; demonstrates significant cost reduction through in-field implementation

- The SCOPES approach (which stands for Situation and/or Problem, Constraints, Objectives and Outcomes, People, Estimates, and Scope), developed by Hwa et al [15], and providing a structured framework to facilitate systematic thinking in design and problem-solving processes, was used to establish a framework for addressing the research problem.
- Three alternative ACD prototypes, based on “Arduino” and “Servo Motor”, were designed by using Design Thinking approach.
- Three alternative ACD prototypes were produced by using 3D printers.
- The software code was developed to operate in harmony with the ICOS system.
- The most suitable ACD alternative for the project was selected by using the WECS (also known as Overall Evaluation Criteria) method.

In this study, the Design Thinking approach was employed for the design of card dispensers. This method was chosen for its ability to prioritize user needs and expectations while fostering innovative and user-centered solutions. Design Thinking is particularly advantageous in understanding real user requirements and developing appropriate solutions, making it a preferred approach in the product development process due to its flexibility and encouragement of creative thinking [18]. Additionally, its emphasis on rapid prototyping and iterative feedback loops helps optimize the design process, ensuring the creation of effective solutions tailored to the project’s requirements.

Design thinking method was introduced by Robert McKim in 1973. It was shaped in a course called “Design Thinking as a Method of Creative Action” given by Rolf Faste at Stanford University between the 1980s and 1990s. This method was used for the first time in business life by IDEO company in 1991 [30]. It is also used by companies such as Apple, Coca-Cola, IBM, Nike, Procter & Gamble and Whirlpool. It identifies the requirements and develops solutions in many areas from the automotive industry to major appliances. Application area varies from the small household appliances to the food industry, and from electronic devices to home appliance design [1, 30, 39]. Among the different approaches to the Design Thinking method, a model including five-stages by Hasso-Plattner Design Institute of Stanford University stands out [5]. The stages of the model are presented in Fig. 4 and described as in the following [8].

Stage 1 – Empathizing: Conducting research to develop an understanding of users. Understanding the problem of the user for whom you are designing.



Fig. 4 Five stages of design thinking

Stage 2 – Defining: Forming a problem statement. Requirements, problems, and insights are identified.

Stage 3 – Ideating: Generating creative solutions to the problem. Ideas and innovative solutions with their assumptions are put forward.

Stage 4- Prototyping: Building a tangible representation of the solution. Prototyping the issues and the problems before producing solutions.

Stage 5 – Testing: Validating the solution with target audience. Testing the proposed solutions.

After designing and developing three alternative ACD prototypes, it is necessary to select the most cost-effective option that best meets the functional requirements. Although the commercially available COTS product offers significantly more features than needed, its price is considerably higher compared to the developed alternatives. Since the goal was to reduce the cost of the card dispenser mechanism and remain within the project budget—and the developed prototypes were capable of fulfilling the minimum functional needs—the selection was made among the three internally developed ACDs. To determine the most suitable alternative, the WECS, one of the multi-criteria decision-making (MCDM) methods, was employed. The main reasons for choosing the WECS among the MCDM techniques are that the project team is the direct user of the product, the presence of time constraints, and the simplicity and rapid applicability of the method. This method allows for an effective and systematic evaluation of different criteria while maintaining an intuitive and comprehensible structure. As a result, a highly feasible decision-making process is provided without time loss or the need for complex mathematical analyses. Other multi-criteria decision-making techniques (e.g., AHP, TOPSIS, ELECTRE or VIKOR) [33] generally require more detailed and complex calculations, extend the decision-making process, and demand additional data and analyses during implementation, making them more costly in terms of time and resources. The overview of the methods used in this study is presented in Fig. 5.

Within the scope of the research, the ethics committee's permission was obtained from the Ethics Committee of Eskisehir Technical University Senate with the decision numbered 16/11 in the meeting held on December 08, 2012. Details regarding the implementation of the Design Thinking approach and the WECS are provided in the following sections.

Designing and developing the ACD alternatives

During the product development phase, a meeting was held with the project team consisting of 1 project coordinator, 6 researchers and 5 scholars. With the brainstorming method, the question of “How might we make a more cost-effective card dispenser instead of the card dispenser planned to be bought from the market within the scope of the project?” was discussed. Hwa et al. [15]’s SCOPES approach was used for a structural framework to solve the problem. The evaluations of our SCOPES approach to the problem are presented in Table 2. SCOPES is the acronym for Situation and/or Problem, Constraints, Objectives and Outcomes, People, Estimates, and Scope.

While designing alternative products, the Design Thinking method was used. In this context, in the empathizing stage; the requirements within the scope of this research

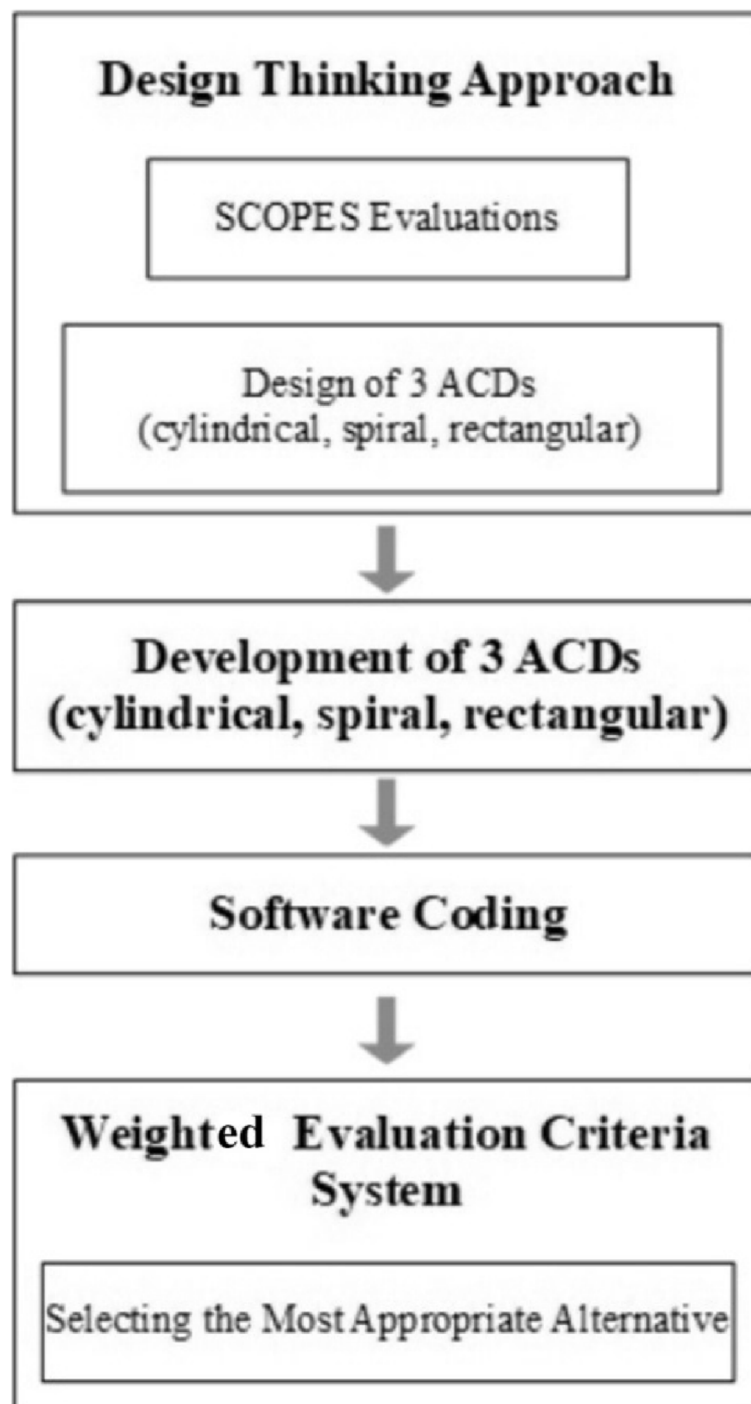


Fig. 5 Overview of the methods used

were investigated. Brainstorming was done with the manager, researchers, and scholars in the project. Card dispenser functions and working principle of the commercial product have been investigated. In the defining stage, the card dispenser to be used in the ICOSs located in the playground is defined as a tool that contains minimum of 40 cards, approximately “85 × 54 mm²” in size. These cards contain pictures and

Table 2 SCOPES evaluations

S Situation and/or Problem	Due to the exchange rate difference, it is necessary to save money and to stay within the project budget. The "Card Dispenser", which is planned to be procured from the market is one of the highest cost items of the HGJ project
C Constraints	The card dispenser restrictions are set as in the following: 1. When the scepter is touched for the first time, an information card should be presented to the participant 2. When a scepter owned by a participant is touched to the same ICOS more than once, only one information card should be given 3. The cost of the card dispenser must be below the cost of the commercially available alternative 4. It should be produced quickly so as not to prolong the delivery time of the project 5. The card holder of the card dispenser must hold a minimum of 40 cards 6. Notifications should be given on the LED screen of the ICOS regarding the situation in ACD
O Objectives and Outcomes	It is aimed to design and produce a card dispenser that can fulfill not all the functions of the card dispenser planned to be procured from the market, but the functions expected from the card dispenser within the scope of the project. Card dispenser must be produced quickly and cost-effectively
P People	Although the aim of the HGJ Project is to provide services to the participants in the playground with ICOSs, the aim of this research is to solve the problem of the project team with an effort to stay within the budget
E Estimates	The existing three-dimensional (3D) printer in the university laboratory, team members who have the competence to produce the card dispenser in software and hardware, the equipment that can be purchased from the market in a cost-effective way to fulfill the desired functions, graduate students who can work within the scope of course projects are the estimated resources that can be used for this research
S Scope	It is possible to produce some products

information about the historical buildings. In the ideating stage, the ACD alternatives that can operate at an optimum level have been produced. At this stage, before moving on to the prototyping stage, the technologies to be used in prototyping are selected. The selected technologies include the use of Arduino card, servo motor and 3D printer. Three different alternatives have been designed. The alternative designs have similar functions but different working principles. The details of the alternative designs are presented below. In the prototyping stage, prototypes of three different designs were produced with the help of a 3D printer. In the testing stage, three alternatives, prototypes of which were produced, were tested according to the determined evaluation criteria. The testing phase is of great importance in determining the best alternative. According to the test results, using the WECS method, three alternatives were compared, and the optimum alternative was selected. Selection of the suitable alternative will be presented below.

Software coding and designing of ACD

The ACD is an apparatus in the ICOSs that will be deployed on the travel route of the playground. This apparatus activates and delivers an information card when the participant touches the ICOS with the scepter. The information card is a card with dimensions of $85 \times 54 \text{ mm}^2$, and thickness of 0.76 mm. It is made up of PVC material and includes a picture and information about the historical building where the ICOS is located. Within the scope of the research, MG995 Tower Pro Brand/Model servo motor that can rotate 360 degrees, has a torque of 10 kg-cm at 4.8 V, can rotate 60 degrees in 0.20 s, and

Arduino Uno microcontroller board are used. Running on the Arduino board, the software/pseudo code, which was written specifically for this project, is presented in Table 3. The software code, which is broadly the same, has been adapted for each alternative.

To support all three card dispenser design alternatives, 360-degree rotational servo motors are employed. The primary challenge was determining the motor's rotation speed and duration for dispensing a single card accurately. The rotation duration was calculated by dividing the required angle to rotate in degrees by the rotation speed of the servo motor in degrees per second. However, experimental studies revealed that environmental conditions such as air temperature, humidity, installation alignments of the servo motor to the chassis, and the amount of card load in the dispenser influenced the required rotation angle. This, in turn, affected the required rotation duration and speed of the servo motor.

To address this, the rotation speed of the servo motor is standardized to 180 degrees per second and then sought to find the optimal rotation duration. Through numerous experiments, it is determined that a nine-degree rotation, just enough to expose the tip of the card with the rectangular storage chamber system that is one of the alternative solution systems explained below, was sufficient for different card loads. This nine-degree rotation required 50 ms of rotation time. However, this duration varied for the other two alternative solution systems named spiral storage chamber and cylindrical storage chamber systems. With the same rotation speed, the spiral storage chamber system required 1200 ms of rotation time, while the cylindrical storage chamber system required 250 ms of rotation time. These durations were determined through extensive experimentation with the dispenser systems. Alternative approaches, such as using microswitches, weight sensors, or optical sensors to detect when the card is dispensed, can be employed to provide feedback to the servo motor to stop the rotation. However, they would increase the production cost for the numerous ICOSs.

According to the form of the card storage chamber, three different alternatives have been designed and modeled namely cylindrical storage chamber system, spiral storage chamber system and rectangular storage chamber system. The COTS product that was

Table 3 Software/pseudo code to run on Arduino board

```
#include <Servo.h> //servo library is included
#define button 12 //button is attached to pin 12 of the Arduino controller
int buton_state = 0; //variable to hold the button state
Servo motor; //motor is defined with the type of Servo
void setup() { //function for initiation
  pinMode (button, INPUT_PULLUP); //button to trigger the rotation
  motor.attach(3); //servo motor is attached to the pin 3 of the controller
}
void loop() { //function for the controller's infinite control loop
  buton_state = digitalRead(button); //read the button state
  if (buton_state == LOW) { //rotation triggered
    motor.write(80); //begin the motor rotation
    delay(50); //rotation duration
    motor.write(90); //stop the rotation
  }
  else //if the rotation is not triggered
    motor.write(90); //stop rotation in an idempotent way,
    delay(300); //sleep to prevent overwhelming the controller
}
```

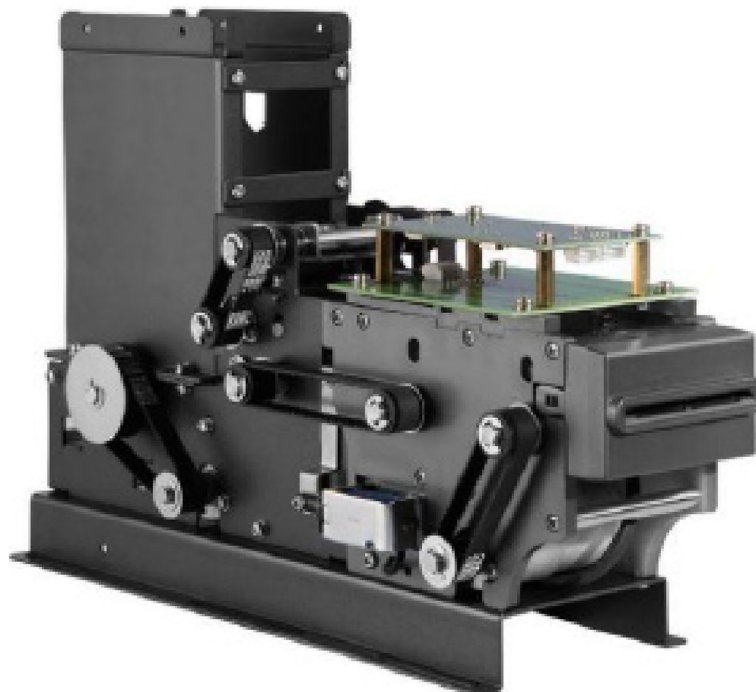


Fig. 6 13.56MHz ISO14443A/B RFID, Mifare card dispenser and reader writer sort mechanism

Table 4 Technical data of COTS product [2]

Subject	Contents
Model	SW-720 K
Operating power	24VDC (max. Current 2 A, static current 0.1A)
Operating condition	Cabin interior
Card sizes	Length: 85 ± 0.5 mm, Width: 54 ± 0.5 mm
Card material	Paper or PVC
Stacker capacity	Standard 150 cards with 0.76 mm thickness
Card type	RFID ISO14443 type A/B
Few cards alert	7 ~ 50pcs (preset 15pcs 0.76 mm thick cards)
Interface	RS232
Dimensions	290*93*204 mm
Weight	2,4 KG

planned to be used at the beginning of the project and the designed alternative ACDs are explained below.

Commercial off-the-shelf (COTS) product

The COTS product which is planned to be supplied within the scope of the project was 13.56MHz ISO14443A/B RFID, Mifare card dispenser and reader writer sort mechanism for parking lot/metro kiosk machine system. A picture of it is presented in Fig. 6.

As of December 2019, the most suitable procurement price from the market is approximately 310 Dollars. Technical data regarding the aforementioned product are presented in Table 4.

The RFID card dispenser operates at 13.56 MHz and supports reading, writing, and sorting of ISO14443A/B-compliant MIFARE cards. Its RS232 interface enables integration with microcontroller-based systems, though it offers lower data rates than USB alternatives. Designed for broad compatibility, the device features a modular structure for easy maintenance, a high-precision transmission channel for accurate card delivery, and jitter card handling to enhance data security. A retrieval box collects invalid or damaged cards, while CNC-level adjustable spacing ensures consistent output. The dispenser also supports functional module customization and features an industrial-grade, corrosion- and abrasion-resistant coating for reliable performance under harsh environmental conditions. These capabilities make the device suitable for various automated systems, including parking lot management, vending machines, bank self-service terminals, telecom and utility card distribution, park ticketing, card initialization and sorting, and public transportation card issuance [2].

Cylindrical storage chamber system

The servo motor is placed right under the middle of the design with the cylindrical card storage chamber, the visuals of which are presented in Fig. 7. When the servo motor is rotated, the cylindrical chamber containing the cards rotates. Mentioned cylindrical chamber rotates on another fixed round plate with a rectangular hole only enough for a card to pass through. For ease of design, it was planned to place 45 information cards inside the cylindrical chamber. For this purpose, slots are opened at 8° intervals for each card on the cylindrical chamber. The cylindrical chamber rotating on another round plate is rotated 8° with the help of the servo motor when the participant comes and touches his/her scepter to the ICOS. At the end of this turn, one card falls into the card presenting chamber from the hole of the round plate on which it moves. The insertion

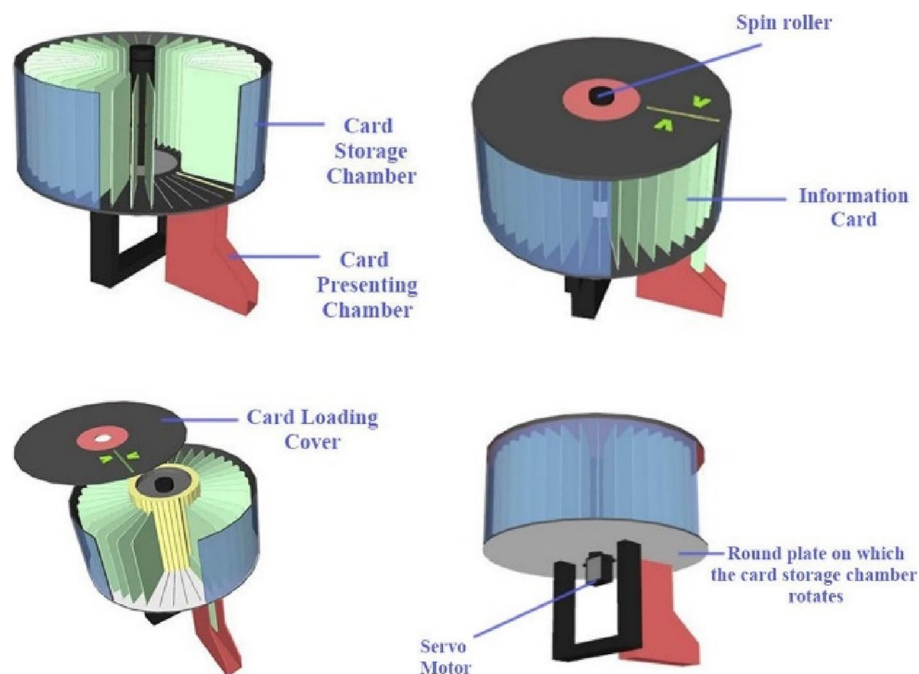


Fig. 7 Model images of cylindrical storage chamber system

of information cards into the emptied card slots is accomplished by opening the circular plate (card loading cover) on top of the storage chamber. While loading the cards into the chamber, one card slot is left empty to prevent the initial card to fall from the open slot. The biggest disadvantage of this design is its lack of flexibility in increasing the number of cards.

Spiral storage chamber system

A spiral is used as the card storage chamber as in Fig. 8. Information cards are placed between the spirals in the card holder. This system is planned to store 40 cards. When the participant comes to ICOS and touches his/her scepter, the servo motor runs and turns the spiral around itself. This movement allows the cards to move towards the open end of the spiral. Thus, the information card at the end falls into the card presentation chamber that is located at the end of the spiral. As the card slots in the spiral are emptied, the cover just on the top of the card storage (card loading cover) can be opened to load cards. The card storage chamber (hence the spiral) can be enlarged as much as the area in the ICOS and the rotating power of the servo motor allow.

Rectangular storage chamber system

Different designs with rectangular storage chambers were produced and the system with rectangular storage chambers was given its final shape according to the evaluation of the advantages and disadvantages of these designs. The visuals of the design are given in

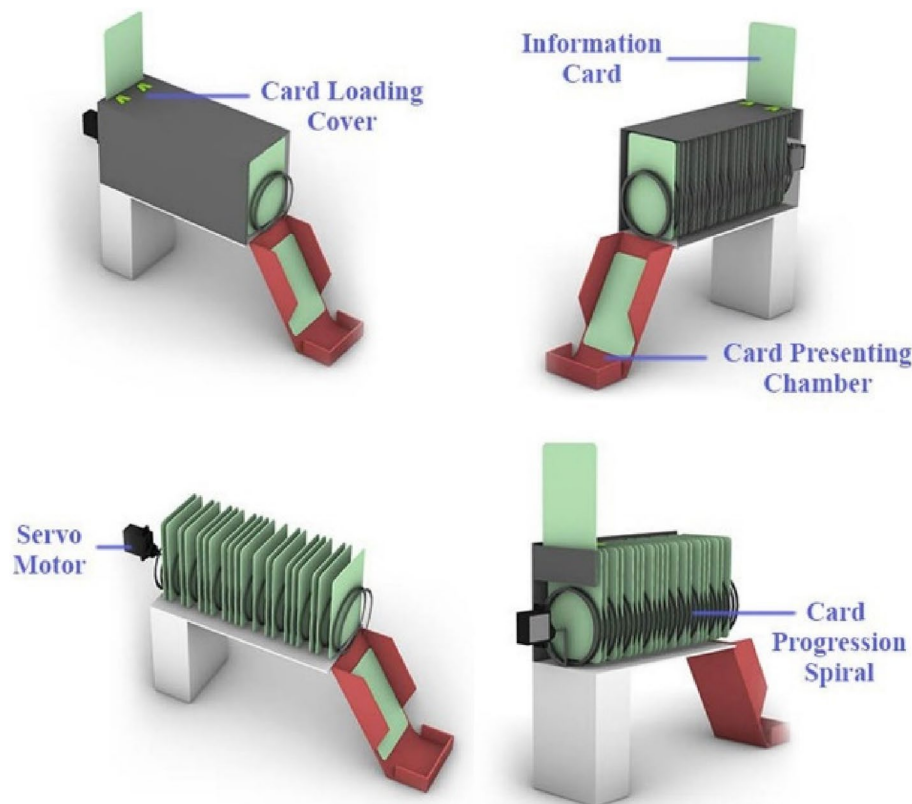


Fig. 8 Model images of spiral storage chamber system

Fig. 9. The cards are placed parallel to the surface and the system is controlled by a rotating cylinder. The bottom surface of the card storage is half open and half closed. A card drop roller connected to the servo motor is placed in the open part. The surface of the roller is covered with a rubber material that can push the cards. When the participant comes to the ICOS and touches his/her scepter, the servo motor connected to the cylinder is rotated and the next card is dragged forward and dropped into the card presentation chamber.

Selection of the suitable alternative

The WECS method is used to select the most appropriate alternative. If a product or process satisfy more than one objective evaluated by different evaluation criteria, it is possible to combine these different evaluation criteria into a single index for the analysis of the overall performance of the product or process [32]. The WECS method is a method in which,

- the examined product or process is evaluated with more than one evaluation criteria,
- each criterion has a different unit of measurement, quality characteristic and relative weight,
- normalization and weighting are done to combine these different criteria,
- more than one product or process can be evaluated using a single numerical index (Weighted Target Score- S_{WT}) created by combining different evaluation criteria [27, 30, 32].

In order to calculate S_{WT} for each alternative, the following processes are carried out:

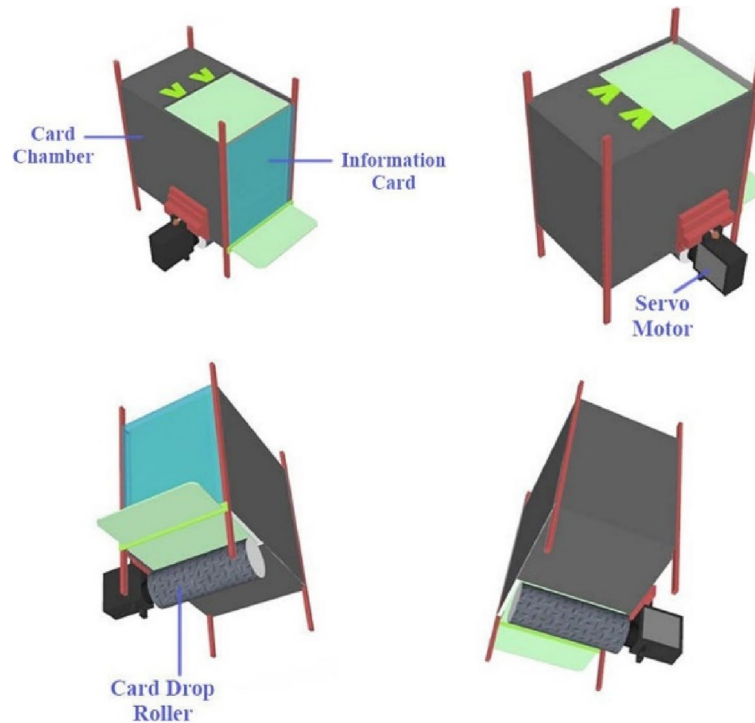


Fig. 9 Model images of rectangular storage chamber system

- Determining evaluation criteria
- Defining the Criteria Weight Value (V_{CW}) of each criterion
- Calculating the Score of Each Alternative According to Each Sub-Criteria (S_{SC})

For determining the evaluation criteria, the studies by Mutlu et al. [27], Parlar et al. [30], and Narayanan et al. [28] were utilized, resulting in the identification of four Main Criteria (MC) and 13 Sub-Criteria (SC), as presented in Fig. 10.

For defining the V_{CW} each criterion, the Automatic Card Dispenser Efficiency Determination Questionnaire (ACDEDEQ), which is included in the Appendix, was applied to HGJ Project team (12 people). While determining the Weight Score (WS) for the MC by the participant, a value should be given for WS of each MC so that when all WSs are added together, the result is 1. Similarly, a value should be given for WS of each SC such that when all WSs related to a MC are added together, the result is 1. WS for each MC and each SC was found by averaging the scores given by all participants. These scores are written in the box under the relevant MC and in the left box under the SC. Criteria Weight Value (V_{CW}) is obtained by multiplying the weight scores of SC and related MC, and is written in the right box below SC. The WSs of all MCs and SCs, and all V_{CW} s are presented in Fig. 10. For calculating the score of each alternative according to each SC, in accordance with the explanations in ACDEDEQ, it is utilized both the predictions of participants (the scores given by the participants for the alternative product are summed and averaged) and the results of tests and calculations.

The Weighted Target Score (S_{WT}) of each alternative product was calculated using the equations below [32]. While finding S_{WT} of an alternative product, a value was found for each sub-criterion by using Equation (1) or Equation (2) and these values were added. S_{WT} found is the score of the alternative product and this score may be maximum 1. The alternative product with the closest score to 1 is the most acceptable alternative. Table 5 presents the Weighted Target Score calculations, with the S_{WT} values for Alternatives 1, 2, and 3 being 0.779, 0.797, and 0.905, respectively. The score of each alternative shows its closeness to the reference (to the ideal situation). The aim here is to reach the most technically appropriate solution, rather than just presenting a solution that fulfills the basic function. Although each of the three alternatives satisfies the stated requirements, Alternative 3 demonstrates the highest level of alignment with them.

Equation (1) is used for the case where the higher value is better.

$$[|S_{SC} - S_W|/|S_B - S_W|] \times V_{CW} \quad (1)$$

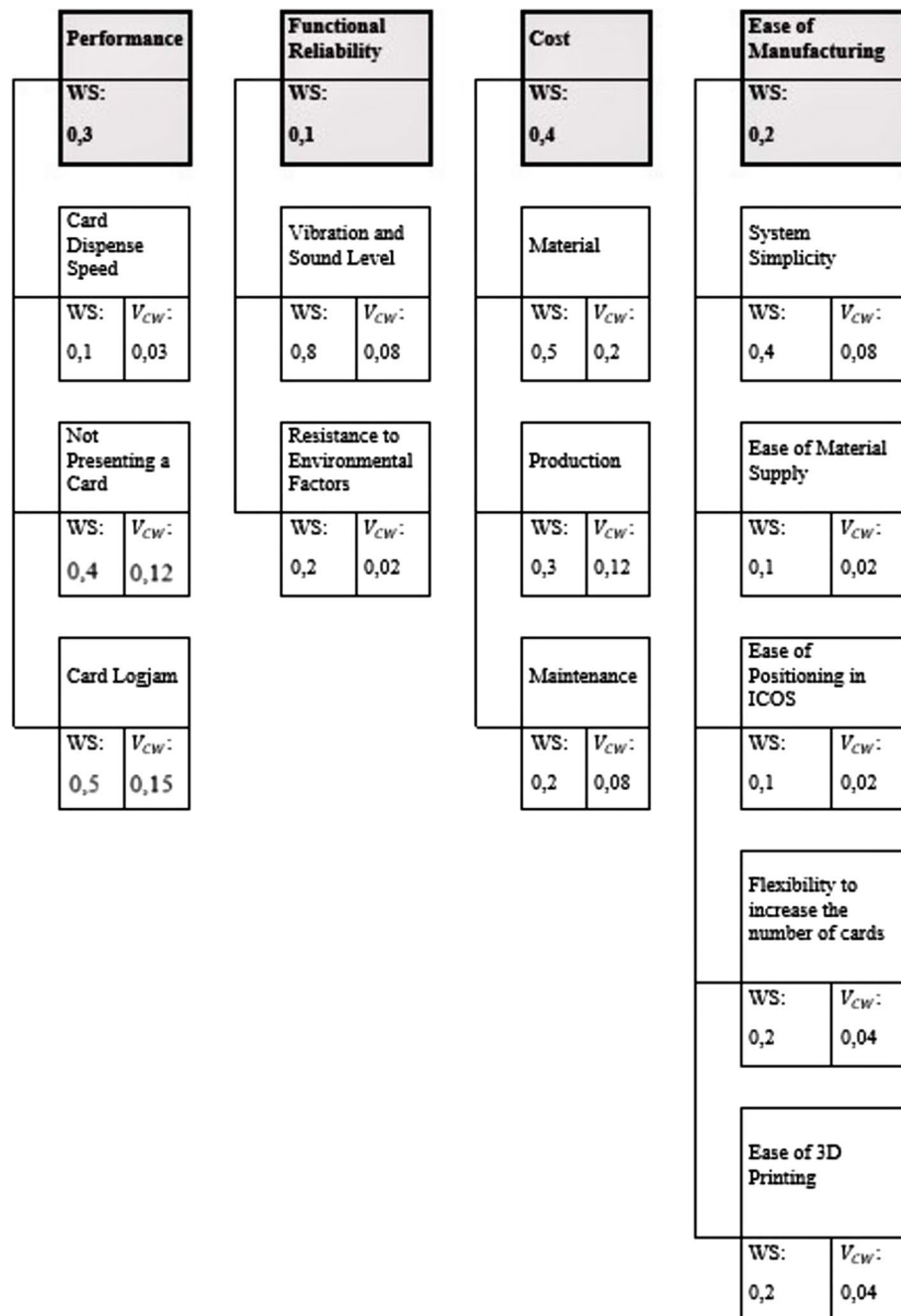
Equation (2) is used for the case where the lower value is better.

$$[1 - (|S_{SC} - S_W|/|S_B - S_W|)] \times V_{CW} \quad (2)$$

“ S_{WT} ”: Weighted Target Score.

“ S_W ”: Weighted Score of SC.

“ S_{SC} ”: Score of each alternative according to each SC.

**Fig. 10** WECS method hierarchy

" S_W ": Worst Score.

" S_B ": Best Score.

" V_{CW} ": Criterion Weight Value.

"S": Lower Value is Better.

"G": Higher Value is Better.

Table 5 Weighted target score calculation table

Criteria	Sub-criteria (SC)	Worst score (S _w)	Best score (S _B)	Quality characteristic	V _{cw}		Cylindrical		Spiral		Rectangular	
							S _{sc}	S _w (F1 or F2)	S _{sc}	S _w (F1 or F2)	S _{sc}	S _w (F1 or F2)
Performance	Card Dispense Speed	10	1	L	0.030	2.000	0.027	0.025	2.500	0.025	1.500	0.028
	Not Presenting a Card	40	0	L	0.120	2.000	0.114	0.111	3.000	0.111	0.000	0.120
Functional Reliability	Card Logjam	40	0	L	0.150	1.000	0.146	0.146	1.000	0.146	0.000	0.150
	Vibration and Sound Level	1	10	H	0.080	7.000	0.053	0.062	8.000	0.062	10.000	0.080
	Resistance to Environmental Factors	1	10	H	0.020	9.000	0.018	0.018	9.000	0.018	9.000	0.018
Cost	Material	100	10	L	0.200	30.000	0.156	0.156	30.000	0.156	30.000	0.156
	Production (Man x Hour)	10	1	L	0.120	2.400	0.101	0.108	1.900	0.108	1.300	0.116
Ease of Manufacturing	Maintenance	10	1	L	0.080	3.240	0.060	0.061	3.190	0.061	3.130	0.061
	System Simplicity	1	10	H	0.080	6.000	0.044	0.036	5.000	0.036	9.000	0.071
	Ease of Material Supply	1	10	H	0.020	9.000	0.018	0.016	8.000	0.016	9.000	0.018
	Ease of Positioning in ICOS	1	10	H	0.020	8.000	0.016	0.018	9.000	0.018	10.000	0.020
	Flexibility to Increase Number of Cards	1	10	H	0.040	2.000	0.004	0.013	4.000	0.013	8.000	0.031
Total (Σ)	Ease of 3D Printing	1	10	H	0.040	6.000	0.022	0.027	7.000	0.027	9.000	0.036
					1.000	0.779			0.797		0.905	

Evaluation

The explanation of the values related to the SC “Card Dispense Speed,” which falls under the MC “Performance” in Table 5, is as follows: Survey participants were asked to assign scores ranging from 10 (S_W) to 1 (S_B) for this criterion, where a lower value indicates better performance (Quality Characteristic). The weight value assigned to this criterion (V_{CW}) is 0.030. For Alternative 1, the score (S_{SC}) calculated during the tests was 2 (second). Since the quality characteristic is classified as type L (“lower value is better”), the weighted score (S_W) for Alternative 1 was calculated as 0.027 using Equation 2. For Alternative 2, the assigned score was 2,5, resulting in a weighted score of 0.025 based on the same equation. For Alternative 3, the score was 1,5, yielding a weighted score of 0.028. The values for other sub-criteria were determined in a similar manner. The total Weighted Target Score (S_{WT}) for Alternative 1 was obtained by summing the S_W values across all sub-criteria. The values for the sub-criteria listed in Table 5—namely, card dispense speed, not presenting a card, card logjam, material cost, production cost, and maintenance cost—were obtained through testing and calculations. The values for the remaining sub-criteria were determined based on the team’s predictions, informed by observations made during the testing process. Detailed explanations for each sub-criterion are provided in the [Appendix](#).

Comparative evaluation

In this study, three design alternatives for the ACD intended for use in the ICOS were evaluated using the WECS. Based on the results of this evaluation, Alternative 3—namely, the Rectangular Storage Chamber System—was identified as the most suitable option for implementation within the HGJ project. Alternative 3 obtained the same weighted score (S_W) as the other two alternatives in the sub-criteria “Resistance to Environmental Factors” and “Material”; received an equal score to Alternative 2 in the “Maintenance” sub-criterion, and Alternative 1 in the “Ease of Material Supply” sub-criterion. In all remaining sub-criteria, such as “Card Dispense Speed,” “Not Presenting a Card,” “Card Logjam,” “System Simplicity,” “Ease of Positioning in ICOS,” “Flexibility to Increase the Number of Cards,” and “Ease of 3D Printing,” Alternative 3 achieved a higher weighted score (S_W) than the other alternatives. Based on the Weighted Target Score (S_{WT}), Alternative 3 achieved a score of 0.905, outperforming Alternative 1 (0.779) and Alternative 2 (0.797). These results indicate that Alternative 3 not only meets the minimum functional requirements but also offers superior performance across critical dimensions, making it the most appropriate solution for the HGJ project. Moreover, the cylindrical storage chamber system (Alternative 1) exhibits a critical drawback: the servo motor fails to consistently achieve the same rotation angle due to card weight and friction. Likewise, the spiral storage chamber system (Alternative 2) suffers from rotational inconsistency caused by the spiral’s mass and card load. In addition, both alternatives offer limited flexibility in expanding card capacity, which further constrains their applicability.

Cost evaluation

The ACD designed, developed, and selected within this study was successfully deployed in the HGJ field implementation, integrated seamlessly with the ICOS system. For the 15 ACD units required, the developed prototype cost \$34.43 per unit, compared to \$310 for COTS product—resulting in a per-unit savings of \$275.57 and a total savings of approximately \$4,134. If COTS products had been used, the ACD component would have accounted for 21.8% of the total project budget. By contrast, the selected prototype reduced this share to just 2.42%, demonstrating substantial financial efficiency and budgetary alignment.

Further planned development

The field-tested success of the developed ACD confirms its practical viability. To further optimize the system, it is planned to utilize the Raspberry Pi controllers already embedded in the ICOS units to perform the functions currently handled by Arduino microcontrollers used in the ACD. This integration will reduce hardware redundancy and enhance cost efficiency. Additional development will focus on improving dispensing accuracy under variable environmental conditions through sensor-based feedback mechanisms. Modular design enhancements are also planned to allow scalable card capacity and simplified maintenance, supporting broader deployment scenarios.

Future recommendations

Although the developed ACD satisfies the essential functional requirements of the HGJ project within a limited budget, further refinement is necessary to elevate it to an industrial-grade product. Future enhancements should focus on incorporating standardized data communication protocols—such as RS232 or USB—to enable seamless plug-and-play integration across diverse system architectures. Expanding the dispenser's capabilities to support multi-card selection and storage would allow it to handle various card types and content categories, thereby increasing its versatility. Additionally, implementing low-card and empty-card alert mechanisms would improve operational continuity by enabling timely maintenance interventions. The integration of card writing functionality, including RFID encoding, could facilitate personalized or trackable card issuance. Finally, adding faulty card detection and separation features would enhance overall system reliability and user experience. These improvements, inspired by the advanced functionalities of COTS products, will contribute to the industrialization of the ACD and support its broader application in domains such as ticketing, gaming, and educational kiosks.

Conclusion

HGJ project was developed to foster cultural heritage awareness among children through gamified experiences delivered via ICOS. Within this system, ACD plays a critical role by distributing information cards upon interaction with a smart scepter.

However, the high cost of COTS dispensers posed a significant budgetary challenge, threatening the feasibility of deploying ICOS units at scale. Therefore, the aim of this study was to design, develop and select a cost-effective ACD prototype that meets the functional requirements of the HGJ project while remaining within its financial constraints.

To achieve this, three ACD prototypes—cylindrical, spiral, and rectangular storage chamber systems—were designed using the Design Thinking approach and the SCOPES framework. Each prototype was developed using Arduino microcontrollers, servo motors, and 3D printing technologies. The most suitable alternative was selected using the WECS, a multi-criteria decision-making method that allowed for systematic comparison across four main-criteria and 13 sub-criteria. The Rectangular Storage Chamber System (Alternative 3) was identified as the optimal solution, achieving the highest Weighted Target Score ($S_{WT}=0.905$) and demonstrating superior performance in key areas such as dispensing reliability, ease of integration, and scalability. Its implementation reduced the ACD cost share in the overall project budget from 21.8% to 2.42%, confirming its financial and functional viability.

The ACD developed within the scope of this study was successfully deployed in the HGJ field implementation, demonstrating both functional reliability and seamless integration with the ICOS system. Its design and original software enabled the project to remain within budget while delivering a cost-effective and versatile dispensing solution. With minor adaptations, the ACD can be practically applied in various domains such as card games, educational kiosks and ticket vending machines used in parking lots, public transportation, highways, ferries, and recreational environments. Although the current prototype meets the operational needs of the HGJ project, further development is required to transform it into an industrial-grade product. Future work will focus on integrating Raspberry Pi controllers to streamline hardware architecture, improving dispensing precision under variable conditions, and enabling modular scalability. It is considered that this study will contribute to the promotion of the TÜBİTAK-supported HGJ project and set an example for similar academic studies with design and informatics content.

Appendix

Automatic Card Dispenser Efficiency Determination Questionnaire (ACDEDEQ)

Chapter 1

1. In the table below, give a score between 1 and 10 for each of the Basic Criteria according to the effect of the Automatic Card Dispenser (ACD) on performing its basic function. The sum of the scores awarded must be 10. (*Provided as an example.)

Performance	Functional reliability	Cost	Ease of manufacturing	Total
3*	1*	4*	2*	10

2. In the table below, give a score between 1 and 10 for each of the Performance Sub-Criteria according to their effect on the performance of ACD. The sum of the scores awarded must be 10.

Card dispense speed	Not presenting a card	Card logjam	Total
			10

3. In the table below, give a score between 1 and 10 for each of the Functional Reliability Sub-Criteria according to their effect on the functional reliability of ACD. The sum of the scores awarded must be 10.

Vibration and sound level	Resistance to environmental factors	Total
		10

4. In the table below, give a score between 1 and 10 for each of the Cost Sub-Criteria according to their effect on the cost of the ACD. The sum of the scores awarded must be 10.

Material	Production (Man x Hour)	Maintenance	Total
			10

5. In the table below, give a score between 1 and 10 for each of Ease of Manufacturing Sub-Criteria according to their effect on ease of manufacture of ACD. The sum of the scores awarded must be 10.

System simplicity	Ease of material supply	Positioning in ICOS	Flexibility to increase the number of cards	Ease of 3D printing	Total
					10

Chapter 2

Give a score between 1 and 10 for each alternative's sub-criteria. Do not enter any values in the fields marked with "X", as these values will be determined through testing and calculations.

Basic Criterion	Sub-Criterion	Explanation	Unit	Alternatives		
				Cylindrical	Spiral	Rectangle
Performance	Card dispense speed	Card dispense speed is the time elapsed between the moment that the scepter touches to ICOS and the moment that the card falls into the card presentation chamber	sec	X	X	X
	Not presenting a card	Not presenting a card is the number of not presenting a card after the scepter touches to ICOS in 40 attempts	piece	X	X	X

Basic Criterion	Sub-Criterion	Explanation	Unit	Alternatives		
				Cylindrical	Spiral	Rectangle
Functional reliability	Card logjam	Card logjam is a situation that requires the operator to intervene because the card cannot be presented both in that attempt and in subsequent attempts due to the logjamming of the card after the scepter touches to ICOS. The number in 40 attempts will be taken	piece (number of interventions)	X	X	X
	Vibration and sound level	Vibration and sound level are rated on a relative scale of 1–10. 10 represents the lowest vibration and sound level, that is, the best condition	1–10			
	Resistance to environmental factors	Resistance to environmental factors is relatively scored between 1–10. 10 represents the most resistant to heat, shaking and rain	1–10			
Cost	Material	Material cost is the total cost of each part and material used in USA dollar	\$	X	X	X
	Production (Man x Hour)	Production cost is calculated as “man x hour” over the time spent by a researcher for code development and the time spent by a scholar during the assembling of materials, drawing the design and printing the prototype from the 3D printer. Total cost is divided by 15 to find the cost of 1 piece	\$	X	X	X
	Maintenance	Maintenance cost is the maintenance cost that the product will need annually. 10% of the material and production cost of the product is taken	\$	X	X	X

Basic Criterion	Sub-Criterion	Explanation	Unit	Alternatives		
				Cylindrical	Spiral	Rectangle
Ease of manufacture	System simplicity	System simplicity is scored on a relative scale of 1–10. 10 represents the simplest, that is, the best, of the system designed considering the difficulties in the design process	1–10			
	Ease of material supply	Ease of material supply is scored on a relative scale of 1–10. 10 represents the easiest way to obtain from the market	1–10			
	Positioning in ICOS	Ease of positioning in ICOS is scored on a relative scale of 1–10. 10 represents the easiest and the best case according to positioning it to ICOS	1–10			
	Flexibility to increase the number of cards	Flexibility to increase the number of cards is scored on a relative scale of 1–10. 10 represents the situation where the number of cards is increased with the least number of adjustments if needed	1–10			
	Ease of 3D printing	Ease of 3D printing is rated on a relative scale of 1–10. 10 represents the situation of putting it together in the easiest way by printing out from a 3D printer	1–10			

Abbreviations

ACD	Automatic card dispenser
ACDEDEQ	Automatic Card Dispenser Efficiency Determination Questionnaire
AHP	Analytic hierarchy process
AM	Additive manufacturing
Atmega328	Advanced Technology Mega 328
CAD	Computer-aided design
COTS	Commercial off-the-shelf
DC	Direct current
ELECTRE	Elimination and Choice Expressing Reality
HGJ	Historical Game Journey
IC	Integrated circuit
ICOS	Information and Communication Station
ICSP	In-Circuit Serial Programming
LED	Light Emitting Diode
MC	Main Criteria

MCDM	Multi-Criteria Decision-Making
mm	Millimeter
RFID	Radio Frequency Identification
SC	Sub-Criteria
SCOPEs	Situation, Constraints, Objectives & Outcomes, People, Estimates, and Scope
THSR	Taiwan High Speed Rail
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
TÜBİTAK	Scientific and Technological Research Institution of Türkiye
USB	Universal Serial Bus
VIKOR	Multi-Criteria Optimization and Compromise Solution
WECS	Weighted Evaluation Criteria System
WS	Weight Score
3D	Three dimensional

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Authors' contributions

HHH contributed to the HGJ project by designing and planning its structure, developing card dispensers, fabricating them via 3D printing, conducting performance tests, coordinating meetings, and overseeing project management. MBKÖ was responsible for the design of card dispensers, material selection and procurement, survey implementation, and manuscript preparation. TÖ focused on dispenser design, material selection and procurement, fabrication, and software development. All authors reviewed and approved the final manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interests in this section.

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